

BazarHQ

SCENARIO DOCUMENT

Multi-Tenant SaaS E-Commerce Platform

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Prepared By	BazarHQ Development Team

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INTRODUCTION TO SCENARIOS

This document provides the complete scenario specification for **BazarHQ** — a multi-tenant SaaS e-commerce platform designed for Bangladeshi merchants. It is derived directly from the BazarHQ Software Requirements Specification (SRS) v1.0, following IEEE 830 standards, and serves as the primary reference for development, testing, and academic evaluation.

What Are Scenarios?

A **scenario** is a narrative description of a specific sequence of events that occurs when an actor interacts with a system to achieve a goal. Scenarios ground abstract requirements in concrete, testable interactions. This document contains three complementary scenario types:

Type	Purpose	Format	Coverage
Use Case Scenarios	Describe actor–system interactions step by step	Happy path + alternative/exception flows	All SRS modules M1–M8, C1–C5, A1–A4
Test Scenarios	Verify functional behaviour under various conditions	Input → Expected Result, typed by category	Positive, Negative, Edge Case, Security, Performance
User Stories	Express requirements from stakeholder perspective	As a [role] I want [goal] so that [benefit] + AC	All stakeholders: Merchant, Customer, Super Admin

System Actors

Actor	Technical Level	Access Point	Primary Modules
Merchant	Low–Medium	{shop}.bazarhq.com/admin	M1–M8
Customer	Low	{shop}.bazarhq.com	C1–C5
Super Admin	High	bazarhq.com/superadmin	A1–A4

SRS Traceability Matrix

Every scenario in this document maps directly to one or more functional requirements in the BazarHQ SRS. The module codes (M1–M8, C1–C5, A1–A4) reference SRS sections. NFRs (performance, security, reliability) are explicitly noted in linked requirements fields of each use case card.

SRS Module	Module Name	Use Cases	Test Cases	User Stories
M1	Registration & Authentication	UC-M01, UC-M02	TC-M01–TC-M08	US-M01
M2	Shop Creation & Onboarding	UC-M03	TC-M09–TC-M10	US-M02
M3	Store Customization	UC-M03 (ext)	TC-M11–TC-M12	US-M03 (ext)
M4	Product Management	UC-M04	TC-M09–TC-M15	US-M02
M5	Order Management	UC-M05	TC-M16–TC-M22	US-M03

M6	Payment Configuration	UC-M06	TC-M23–TC-M27	US-M04
M7	Analytics & Reporting	UC-M07	TC-M28–TC-M32	US-M05
M8	Account Settings	UC-M08 (ext)	TC-M33–TC-M36	US-M06
C1	Shop Discovery & Browsing	UC-C01	TC-C01–TC-C04	US-C01
C2	Cart Management	UC-C02 (cart)	TC-C05–TC-C08	US-C01
C3	Checkout & Payment	UC-C02	TC-C09–TC-C14	US-C01, US-C02
C4	Order Tracking	UC-C03	TC-C15–TC-C18	US-C02
C5	Customer Account	UC-C04 (ext)	TC-C19–TC-C22	US-C03
A1	Admin Authentication	UC-A01	TC-A01–TC-A08	US-A01
A2	Merchant Management	UC-A02	TC-A09–TC-A14	US-A02
A3	Platform Analytics	UC-A03	TC-A15–TC-A20	US-A03
A4	Theme & Content Mgmt	UC-A04 (ext)	TC-A21–TC-A24	US-A04

PART 1 — USE CASE SCENARIOS

Use case scenarios describe how each actor interacts with BazarHQ to achieve a specific goal. Each use case includes a unique ID, involved actors, preconditions, main (happy) path, alternative/exception flows, and traceability to SRS functional requirements.

1.1 Merchant Use Cases

UC-M01 Merchant Registration	
Actors: Merchant, BazarHQ Authentication System	Description: A new merchant creates an account using email/password or Google OAuth.
Preconditions • Merchant does not have an existing account. • Registration page is accessible.	Postconditions ✓ Merchant account is created. ✓ Verification email is sent. ✓ Merchant is prompted to verify email before dashboard access.
Step	Main Flow — Happy Path
1	Merchant navigates to bazarhq.com and clicks 'Sign Up'.
2	Merchant enters full name, email address, and password (min 8 chars, 1 number).
3	System validates that the email is unique and password meets strength requirements.
4	System creates the account and sends a verification email.
5	Merchant opens the email and clicks the verification link.
6	System marks the account as verified and prompts the merchant to create a shop.
Alt #	Alternative / Exception Flows
A1	Email already exists → System displays 'Email already registered' and prompts login.
A2	Password too weak → System highlights the violated rule before form submission.
A3	Merchant chooses Google OAuth → Redirects to Google, retrieves profile, creates account, skips email verification.
A4	Verification email not received → Merchant can request resend from the login page.
Linked FRs: M1-FR: Register via email/OAuth, send verification email, validate email uniqueness, enforce password strength.	Linked NFRs: M1-NFR: bcrypt password hashing, login ≤2s, HTTPS, failed attempt logging.

UC-M02 Merchant Login	
Actors: Merchant, BazarHQ Authentication System	Description: A registered merchant authenticates and gains access to their dashboard.
Preconditions • Merchant has a verified account. • Login page is accessible.	Postconditions ✓ Merchant is redirected to their dashboard. ✓ JWT session (7-day expiry) is created.
Step	Main Flow — Happy Path
1	Merchant navigates to {shop}.bazarhq.com/admin and is redirected to login page.
2	Merchant enters registered email and password.

3	System validates credentials against the hashed password in the database.	
4	System issues a JWT token and redirects the merchant to the dashboard.	
Alt #	Alternative / Exception Flows	
A1	Wrong credentials → System displays 'Invalid email or password'; attempt counter incremented.	
A2	5 consecutive failures → Account locked temporarily; merchant notified via email.	
A3	Suspended account → Merchant sees a suspension message with the stated reason, not a generic error.	
A4	Google OAuth login → Merchant clicks 'Login with Google'; system authenticates via OAuth and creates session.	
Linked FRs: M1-FR: Login via email/OAuth, redirect on success, display error on failure, lock after 5 failures.		Linked NFRs: M1-NFR: Session JWT 7-day expiry, login response ≤2s, HTTPS encryption.

UC-M03 Shop Creation & Subdomain Assignment		
Actors: Merchant, BazarHQ Shop Management System		Description: A verified merchant creates their online shop, selects a unique subdomain, and begins onboarding.
Preconditions • Merchant is logged in and email is verified. • Merchant has no existing shop (V1: 1 shop per account).		Postconditions ✓ Shop is created in draft status. ✓ Unique subdomain is reserved. ✓ Merchant enters the onboarding checklist.
Step	Main Flow — Happy Path	
1	System prompts merchant to create a shop immediately after email verification.	
2	Merchant enters a shop name.	
3	System auto-generates a subdomain suggestion (e.g. rahims-leather.bazarhq.com).	
4	System performs a real-time availability check (responds within 1 second).	
5	Merchant confirms or edits the subdomain (lowercase letters, numbers, hyphens only).	
6	Merchant submits; system saves the shop as Draft and redirects to the onboarding checklist.	
Alt #	Alternative / Exception Flows	
A1	Subdomain taken → System immediately flags conflict and suggests alternatives.	
A2	Reserved word used (admin, api, www, etc.) → System rejects and explains restriction.	
A3	Browser closed mid-flow → Progress is saved; merchant resumes on next login.	
A4	Merchant attempts to create a second shop → System blocks with a V1 limitation message.	
Linked FRs: M2-FR: Prompt shop creation after verification, auto-generate subdomain, real-time availability check, save as Draft.		Linked NFRs: M2-NFR: Subdomain check ≤1s, shop creation ≤3s, unique DB constraint, data isolation.

UC-M04 Add Product with Variants		
Actors: Merchant, Cloudinary, BazarHQ Inventory System		Description: Merchant adds a new product including variant definitions, images, and stock levels.
Preconditions • Merchant is logged in. • Shop exists. • Cloudinary storage is operational.		Postconditions ✓ Product is saved (Active or Draft). ✓ Stock counts recorded per variant. ✓ Images stored on Cloudinary via CDN.
Step	Main Flow — Happy Path	

1	Merchant navigates to Products → Add New Product.
2	Merchant enters product name, description, base price, and category.
3	Merchant uploads 1–6 product images (max 2MB each); system compresses and stores via Cloudinary.
4	Merchant designates one image as the primary thumbnail.
5	Merchant adds variant types (e.g. Size: S, M, L; Color: Red, Blue) with individual stock quantities and prices.
6	Merchant sets product status to Active or Draft and clicks Save.
7	System records the product and updates stock count in real time.
Alt #	Alternative / Exception Flows
A1	Image exceeds 2MB → System blocks upload and shows a clear error message.
A2	Missing mandatory field (name or price) → System prevents saving and highlights missing fields.
A3	A variant goes out of stock later → System marks that variant unavailable; product remains visible.
A4	Merchant duplicates an existing product → System creates a copy with all fields pre-filled for editing.
Linked FRs: M4-FR: Add product with name/description/price/category/stock, upload 1–6 images, define variants, set Active/Draft status.	
Linked NFRs: M4-NFR: Images auto-compressed via CDN, product page ≤3s, CSV import 500 products ≤30s, real-time stock update.	

UC-M05 Order Management — Receive & Fulfil Order	
Actors: Merchant, Customer, BazarHQ Notification System, SMS/Email Gateway	
Description: Merchant receives a new order notification and progresses it through the order lifecycle.	
Preconditions • Merchant is logged in. • At least one active product exists. • Customer has placed an order.	Postconditions ✓ Order status is updated. ✓ Customer notified at each status change. ✓ Invoice can be downloaded.
Step	Main Flow — Happy Path
1	Customer places an order; system triggers email and SMS notification to the merchant.
2	A real-time badge appears on the merchant dashboard indicating a new order.
3	Merchant opens the Order List and clicks the new order.
4	Order Detail page shows: customer info, items, payment method, address, and order timeline.
5	Merchant reviews and clicks 'Confirm Order' → Status changes to Confirmed.
6	Merchant arranges shipment and clicks 'Mark as Shipped' → Status changes to Shipped.
7	Upon delivery confirmation, merchant clicks 'Mark as Delivered' → Status changes to Delivered.
8	Customer receives an SMS/email notification at each status change.
9	Merchant downloads a PDF invoice from the order detail page.
Alt #	Alternative / Exception Flows
A1	SMS delivery fails → System automatically falls back to email notification.
A2	Merchant cancels order → Must provide a cancellation reason; customer is notified.
A3	Order delivered → Cannot be cancelled through the platform; merchant must resolve directly with customer.
A4	COD order → Payment status remains 'Pending' until merchant manually marks it as 'Collected'.

A5	No action for 48 hours on Pending order → System sends a reminder notification to the merchant.
Linked FRs: M5-FR: Email+SMS on new order, real-time dashboard badge, order detail view, status lifecycle Pending→Confirmed→Shipped→Delivered, PDF invoice.	Linked NFRs: M5-NFR: Order list ≤2s with 1000+ records, notifications ≤30s, status update reflected on tracking ≤5s, no orders lost.

UC-M06 Configure Payment Methods	
Actors: Merchant, SSLCommerz, bKash, Nagad, BazarHQ Payment Module	Description: Merchant enables and configures payment methods available to customers at checkout.
Preconditions • Merchant is logged in. • Shop exists. • Merchant holds valid gateway credentials.	Postconditions ✓ Selected payment methods are active. ✓ Credentials are encrypted and stored. ✓ Customers see only enabled methods at checkout.
Step	Main Flow — Happy Path
1	Merchant navigates to Settings → Payment Configuration.
2	Merchant enables desired payment methods (bKash, Nagad, SSLCommerz, COD, Rocket).
3	For bKash/Nagad: Merchant enters their merchant number.
4	For SSLCommerz: Merchant enters Store ID and Store Password.
5	System validates credential format and saves; sensitive fields are masked (last 4 chars shown).
6	Merchant publishes shop; customers now see only the enabled payment options at checkout.
Alt #	Alternative / Exception Flows
A1	Invalid credential format → System rejects before saving; highlights the invalid field.
A2	SSLCommerz credentials wrong → System disables card payments and notifies merchant; raw gateway errors not exposed.
A3	All payment methods disabled → System prevents shop from being published until at least one is active.
A4	Merchant disables a method → Previous orders using that method remain unaffected.
Linked FRs: M6-FR: Enable/disable bKash, Nagad, Rocket, SSLCommerz, COD; mask credentials; validate format; require at least one active method.	Linked NFRs: M6-NFR: Credentials stored encrypted, never stored in plain text, BazarHQ never stores card numbers, all payment API calls over HTTPS.

UC-M07 View Analytics Dashboard	
Actors: Merchant, BazarHQ Analytics Engine	Description: Merchant views revenue, order, product, and visitor analytics to monitor shop performance.
Preconditions • Merchant is logged in. • Shop has at least some order or visitor data.	Postconditions ✓ Dashboard displays up-to-date analytics (refreshed within 5 minutes). ✓ Merchant can export CSV reports.
Step	Main Flow — Happy Path
1	Merchant navigates to Analytics from the dashboard sidebar.
2	System displays revenue totals (day/week/month) and a 30-day revenue trend chart.
3	System displays order count, breakdown by status, and average order value.
4	System displays top-selling products by units sold and by revenue.
5	System displays visitor count, conversion rate, and most-visited pages.

6	Merchant selects a date range and clicks 'Export CSV' to download an order or revenue report.	
Alt #	Alternative / Exception Flows	
A1	Zero orders → Dashboard shows empty state illustrations with setup guidance instead of blank graphs.	
A2	Analytics service slow → Dashboard loads within 3 seconds; stale data shown with 'Last updated' timestamp.	
A3	Cancelled orders → Excluded from revenue totals but counted in order stats with 'Cancelled' label.	
Linked FRs: M7-FR: Revenue totals (day/week/month), 30-day trend chart, order breakdown by status, top products, visitor analytics, CSV export.		Linked NFRs: M7-NFR: Dashboard loads ≤3s, analytics updated ≤5 min, data retained ≥12 months, read-only.

UC-M08 Account Settings & Security		
Actors: Merchant, BazarHQ Authentication System, Email/SMS Gateway		Description: Merchant manages their profile, security settings, notification preferences, and account lifecycle.
Preconditions • Merchant is logged in. • Account exists and is active.		Postconditions ✓ Profile/security changes are saved. ✓ Relevant confirmations and notifications are sent.
Step	Main Flow — Happy Path	
1	Merchant navigates to Account Settings.	
2	Merchant updates full name, contact phone number, or profile picture.	
3	Merchant initiates email change; system sends verification to new email before updating.	
4	Merchant changes password by entering current and new password.	
5	Merchant enables 2FA via authenticator app (TOTP/RFC 6238).	
6	Merchant views active sessions and remotely terminates any session.	
7	Merchant configures notification preferences (email / SMS / both; toggle notification types).	
Alt #	Alternative / Exception Flows	
A1	Google OAuth account → 'Change Password' option hidden with explanatory note.	
A2	Email change → New email must be verified before taking effect; old email remains active until then.	
A3	Account deletion requested with active orders → System warns merchant to fulfil orders first.	
A4	Lost 2FA app → Recovery only via backup codes provided during 2FA setup.	
A5	Account deletion → Shop unpublished immediately; full deletion completed within 30 days.	
Linked FRs: M8-FR: Update name/email/phone/photo, change/reset password, enable 2FA, manage sessions, set notification preferences, delete account.		Linked NFRs: M8-NFR: Password reset link expires in 1hr, TOTP RFC 6238 compliant, setting changes confirmed ≤2s, deletion irreversible after 30-day grace.

1.2 Customer Use Cases

UC-C01 Browse Products & Search

Actors: Customer, BazarHQ Storefront Engine

Description: A customer visits a merchant's storefront, browses the catalogue, and searches for items.

Preconditions

- Merchant's shop is published.
- Customer has a web browser and internet connection.
- No account required.

Postconditions

- ✓ Customer has found desired products and can proceed to add them to the cart.

Step	Main Flow — Happy Path
1	Customer navigates to {shop}.bazarhq.com via a direct link or shared URL.
2	System loads the storefront in the merchant's selected theme within 3 seconds.
3	Customer browses homepage — hero banner, featured products, and category navigation are visible.
4	Customer clicks a product category to filter the catalogue.
5	Customer applies price range filter and sorts by 'Price: Low to High'.
6	Customer clicks a product card to open the Product Detail Page.
7	Detail page shows image gallery, description, price, variants, and stock status.
8	Customer selects a variant (e.g. Size: M, Color: Blue); price and stock update dynamically.
9	Customer clicks 'Add to Cart'.
Alt #	Alternative / Exception Flows
A1	Customer visits non-existent subdomain → System shows a friendly 'Shop Not Found' page.
A2	Shop is unpublished → System shows 'Shop Currently Unavailable'; no shop data is revealed.
A3	Search yields no results → System shows 'No products found' with a suggestion to browse categories.
A4	Out-of-stock product → Card is clearly marked; 'Add to Cart' button is disabled.
A5	Stock < 5 units → Product detail page shows 'Only X left in stock' warning.
Linked FRs: C1-FR: Guest access via subdomain, theme rendering, category browse, price filter/sort, product detail page, variant selection, stock status. Linked NFRs: C1-NFR: Homepage ≤3s on mobile, search results ≤1s, fully responsive from 320px, CDN image delivery, works on low-end Android.	

UC-C02 Cart Management & Checkout

Actors: Customer, BazarHQ Cart System, BazarHQ Order System, Payment Gateways

Description: Customer builds a cart and completes a 3-step checkout process to place an order.

Preconditions

- Customer has at least one item in the cart.
- Merchant has at least one payment method active.

Postconditions

- ✓ Order is placed.
- ✓ Cart is cleared.
- ✓ Customer receives confirmation SMS (and email if provided).
- ✓ Merchant receives order notification.

Step	Main Flow — Happy Path
1	Customer opens the cart and reviews items — image, variant, unit price, quantity, and line totals are shown.

2	Customer adjusts quantity if needed; subtotal, delivery charge, and total update instantly.
3	Customer clicks 'Proceed to Checkout'.
4	Step 1 — Delivery Info: Customer enters full name, phone (01XXXXXXXXXX), full address, and district.
5	Step 2 — Payment: Customer selects from available payment methods (e.g. bKash).
6	For bKash/Nagad: Customer completes payment on their phone and enters the Transaction ID.
7	For SSLCommerz/Card: Customer is redirected to the payment gateway and returns to BazarHQ after completion.
8	Step 3 — Review: Customer reviews the full order summary including policies.
9	Customer clicks 'Place Order'; system creates the order and shows a success page with Order ID.
10	System sends order confirmation SMS and email (if provided) within 30 seconds.
Alt #	Alternative / Exception Flows
A1	Item goes out of stock between cart and checkout → System alerts customer; checkout blocked until resolved.
A2	Price changed after adding to cart → System updates cart and displays a notice before checkout.
A3	SSLCommerz unavailable → System shows a clear error and suggests alternative payment methods.
A4	bKash/Nagad orders → Status is 'Payment Pending Verification' until merchant confirms.
A5	Session expires mid-checkout → Cart is preserved; customer returns to Step 1.
Linked FRs: C2-FR: Add/remove/modify cart items, 3-step checkout without account, phone validation (01XXXXXXXXXX), payment method selection, order confirmation. Linked NFRs: C2-NFR: Full checkout ≤3 min for first-time mobile user, checkout pages ≤2s, payment data never stored on BazarHQ, simultaneous checkout conflict-free.	

UC-C03 Order Tracking	
Actors: Customer, Merchant, BazarHQ Order System, SMS Gateway	Description: Customer tracks the progress of a placed order without requiring an account.
Preconditions • Customer has placed an order. • Customer has their Order ID and registered phone number.	Postconditions ✓ Customer sees current order status and full timeline. ✓ Customer notified via SMS on each status update.
Step	Main Flow — Happy Path
1	Customer clicks the tracking link from their order confirmation SMS or navigates to {shop}.bazarhq.com/track.
2	Customer enters their Order ID and phone number.
3	System validates the details and displays the order tracking page.
4	Visual timeline shows: Order Placed → Payment Confirmed → Order Confirmed → Shipped → Delivered.
5	Completed stages are highlighted; the current active stage is clearly indicated with timestamp.
6	Order summary section shows items ordered, total, payment method, and delivery address.
7	Merchant's shop name and contact information are visible for the customer to reach out if needed.
Alt #	Alternative / Exception Flows
A1	Incorrect Order ID or phone number → System shows 'Order not found'; does not reveal whether the Order ID exists.
A2	No status update for 48+ hours → System shows last known status without implying delay.
A3	Order cancelled → Tracking page shows 'Cancelled' with the merchant's stated cancellation reason.
A4	Customer opts out of SMS → System records preference; no further SMS notifications are sent.

A5	Order delivered 90+ days ago → Tracking page is archived; message explains the record is no longer active.
Linked FRs: C4-FR: Tracking page at /track, look up by Order ID + phone, visual timeline with timestamps, SMS on each status update, opt-out of SMS.	Linked NFRs: C4-NFR: Tracking page ≤2s, status reflects merchant update ≤5s, accessible on basic mobile browsers, SMS ≤30s.

UC-C04 Customer Account (Optional)	
Actors: Customer, BazarHQ Authentication System	Description: A customer creates an optional account to access order history, saved addresses, and faster checkout.
Preconditions - Customer wishes to create an account. - Customer is on a published BazarHQ storefront.	Postconditions ✓ Account is created. ✓ Customer can view order history and save up to 3 delivery addresses.
Step	Main Flow — Happy Path
1	Customer clicks 'Create Account' on any BazarHQ storefront or after placing a guest order.
2	Customer enters email and password, or registers via Google OAuth.
3	System sends a verification email.
4	Customer verifies email and logs in.
5	Customer views full order history across all BazarHQ shops.
6	Customer saves up to 3 delivery addresses; sets one as default.
7	On next checkout, default address is pre-filled in Step 1.
Alt #	Alternative / Exception Flows
A1	Guest customer places order with email already registered → System associates order with that account and notifies customer.
A2	Customer deletes account → Loses access to order history on platform; merchants retain order data.
A3	Google OAuth account → 'Change Password' option hidden from account settings.
A4	3 addresses already saved → System informs limit reached; prompts to delete an existing address first.
Linked FRs: C5-FR: Email/OAuth registration, order history across shops, save up to 3 addresses, set default address, update name/phone, delete account.	Linked NFRs: C5-NFR: Customer data separate from merchant data at DB level, bcrypt password hashing, session tokens expire after 30 days inactivity.

1.3 Super Admin Use Cases

UC-A01 Super Admin Login with 2FA

Actors: Super Admin, BazarHQ Authentication System, Authenticator App

Description: A BazarHQ team member securely authenticates into the Super Admin panel using email, password, and TOTP 2FA.

Preconditions

- Admin account exists (created at DB level).
- Admin has a configured 2FA authenticator app.
- IP address is whitelisted.

Postconditions

- ✓ Admin is logged in with an 8-hour session (30-minute inactivity timeout).
- ✓ All authentication events written to immutable audit log.

Step	Main Flow — Happy Path
1	Admin navigates to bazarhq.com/superadmin (access denied for non-whitelisted IPs with no technical details shown).
2	Admin enters email and password.
3	System validates credentials; if correct, prompts for 2FA code.
4	Admin opens their authenticator app and enters the current TOTP code.
5	System validates the TOTP code against RFC 6238 standard.
6	Admin is redirected to the platform overview dashboard.
7	All authentication events are written to the immutable audit log.
Alt #	Alternative / Exception Flows
A1	Non-whitelisted IP → System shows a blocked access page with no technical details.
A2	Wrong password → Attempt counter incremented; lockout after 3 consecutive failures for 30 minutes; email alert sent.
A3	Expired or incorrect TOTP → Access denied; admin must wait for the next time-window code.
A4	Lost authenticator app → Account recovery only via internal developer-level process.
A5	Inactivity for 30 minutes → Session auto-terminates; admin must re-authenticate.
Linked FRs: A1-FR: Separate admin login at /superadmin, 2FA TOTP challenge, IP whitelist, failed login lockout after 3 attempts, 8hr session/30min inactivity, audit log. Linked NFRs: A1-NFR: HTTPS always, admin page not indexed by search engines, httpOnly cookies for session tokens, audit log immutable, TOTP RFC 6238 compliant.	

UC-A02 Merchant Account Suspension

Actors: Super Admin, Merchant, BazarHQ Notification System

Description: Super Admin suspends a merchant account due to policy violation, immediately revoking access and unpublishing their shop.

Preconditions

- Admin is logged in.
- Target merchant account exists and is currently active.

Postconditions

- ✓ Merchant logged out of all sessions.
- ✓ Shop is unpublished.
- ✓ Merchant receives suspension email.
- ✓ Action recorded in audit log.

Step	Main Flow — Happy Path
1	Admin navigates to Merchant Management and searches for the target merchant by name or email.
2	Admin opens the merchant's detail page — reviews account info, order history, and activity log.
3	Admin clicks 'Suspend Account'.

4	System requires admin to enter a suspension reason before proceeding.
5	Admin enters the reason and confirms.
6	System immediately terminates all of the merchant's active sessions.
7	System unpublishes the merchant's shop.
8	System sends an email to the merchant with the suspension reason.
9	Action is recorded in the immutable audit log with admin ID, timestamp, and IP address.
Alt #	Alternative / Exception Flows
A1	Merchant has active pending orders → System warns admin; those orders remain visible on customer tracking but merchant cannot log in to fulfil them.
A2	Admin cancels mid-flow → No action is taken; merchant remains active.
A3	Admin reinstates merchant → System sends a reinstatement email; merchant can log in and republish their shop.
A4	Admin tries to view payment credentials → System blocks; credentials are encrypted and inaccessible even at admin level.
Linked FRs: A2-FR: Suspend/reinstate merchant, require suspension reason, terminate all sessions, unpublish shop, send suspension/reinstatement email, log all actions.	
Linked NFRs: A2-NFR: Suspension takes effect ≤5s, all merchant management actions written to immutable audit log immediately, permanent deletion processed ≤30 days.	

UC-A03 Platform Analytics & System Health Monitoring	
Actors: Super Admin, BazarHQ Analytics Engine, System Health Monitor	Description: Super Admin monitors platform-wide KPIs, merchant activity, and operational status of all critical services.
Preconditions - Admin is logged in. - Platform has registered merchants and order data.	Postconditions ✓ Admin has a real-time view of platform health. ✓ Admin can export analytics reports.
Step	Main Flow — Happy Path
1	Admin lands on the platform overview dashboard after login.
2	Dashboard shows in real time: total merchants, active shops, orders today/month, and platform revenue this month.
3	Admin reviews 30-day graphs for new merchant registrations and total order volume.
4	Admin clicks Merchant Analytics to see the top 10 highest-revenue merchants and registration trends.
5	Admin clicks Order & Revenue Analytics to see order breakdown by status and revenue by payment method.
6	Admin selects a custom date range and exports a full analytics CSV report.
7	Admin navigates to System Health to see status of web server, database, storage, email, and SMS gateways.
8	Admin reviews server response time, error rate (last 24 hours), and the error log (last 7 days).
Alt #	Alternative / Exception Flows
A1	Critical service goes down → System sends email alert to admin; health monitor reflects outage within 2 minutes.
A2	Export for large date range → Processed as a background job; delivered to admin via email when ready.
A3	Admin tries to modify analytics data → All analytics views are read-only; no edit functionality is available.
A4	Deleted merchant's historical data → Still included in platform revenue aggregates; individual record is removed.
Linked FRs: A3-FR: Platform overview dashboard, merchant analytics, order/revenue analytics by payment method, custom date range, CSV export, system health monitor, audit log.	
Linked NFRs: A3-NFR: Overview dashboard ≤3s, analytics aggregated every 15 min, health monitor checks every 60s with ≤2 min update, audit log retained ≥2 years.	

UC-A04 Theme & Content Management	
Actors: Super Admin, BazarHQ Theme Engine, All Merchant Storefronts	Description: Super Admin manages available storefront themes and controls platform-level content such as announcements and policy pages.
Preconditions - Admin is logged in. - Platform has themes available and merchant storefronts active.	Postconditions ✓ Themes are updated or added. ✓ Merchants receive notifications of changes. ✓ Platform content is updated within 60 seconds.
Step	Main Flow — Happy Path
1	Admin navigates to Theme Management and reviews the list of all available themes.
2	Admin previews a theme before making it available or setting it as default for new shops.
3	Admin activates or deactivates a theme; informs merchants of upcoming deprecations via email.
4	Admin navigates to Announcements and composes a broadcast message to all merchants.
5	Admin schedules or immediately sends the announcement; it appears in merchant dashboards and via email.
6	Admin navigates to Platform Content and updates the Terms of Service or Privacy Policy.
7	Second super admin approves the policy update; it goes live with an updated date stamp.
Alt #	Alternative / Exception Flows
A1	Admin attempts to delete a theme in use → System blocks; explains which merchants are using it.
A2	Broadcast announcement already sent → Cannot be recalled; confirmation screen shown before sending.
A3	Theme update has breaking changes → Admin must notify affected merchants at least 7 days in advance.
A4	Policy update requires second approval → System blocks publication until a second admin approves.
Linked FRs: A4-FR: Theme list/preview/add/update/activate/deactivate/default, broadcast announcements, schedule announcements, update ToS/Privacy Policy with dual approval, FAQ management.	Linked NFRs: A4-NFR: Theme preview ≤3s, announcements delivered to all merchants ≤10 min, platform content updates live ≤60s, theme assets served via CDN.

PART 2 — TEST SCENARIOS (TEST CASES)

Test scenarios verify that BazarHQ's functional and non-functional requirements behave correctly under positive, negative, security, edge-case, and performance conditions. Each test case specifies inputs/actions and the expected system response.

Legend: ● Positive ● Negative ● Edge Case ● Security ● Performance

2.1 Merchant Tests

2.1.1 Registration & Authentication (SRS M1)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-M01	Valid registration with email/password	Email: new@shop.com, Password: Shop1234	Account created; verification email sent; success message shown.	Positive
TC-M02	Duplicate email registration	Email already in system, Password: Shop1234	Error: 'Email already registered'. No new account created.	Negative
TC-M03	Weak password — no number	Email: new@shop.com, Password: shoppass	Form rejected with password strength error before submission.	Negative
TC-M04	Google OAuth registration	Click 'Register with Google', select Google account	Account created via OAuth; email verification skipped; shop creation prompted.	Positive
TC-M05	Login with correct credentials	Registered email and correct password	JWT session created; merchant redirected to dashboard.	Positive
TC-M06	Login with wrong password x5	Correct email, wrong password submitted 5 times	Account locked temporarily after 5th attempt; email notification sent to merchant.	Negative
TC-M07	Suspended account login attempt	Email and password of a suspended account	Suspension message with stated reason displayed; not a generic error.	Edge Case
TC-M08	Dashboard access without email verification	Unverified account attempts to access dashboard URL directly	Redirect to email verification prompt; dashboard inaccessible.	Security

2.1.2 Shop Creation & Onboarding (SRS M2)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-M09	Create shop with valid subdomain	Shop name: 'Rahim Leather', subdomain: rahims-leather	Shop created as Draft; subdomain rahims-leather.bazarhq.com reserved; onboarding checklist shown.	Positive
TC-M10	Subdomain already taken	Enter subdomain already in use by another merchant	System immediately flags conflict; suggests alternatives in real time.	Negative
TC-M11	Reserved word as subdomain	Subdomain: admin	System rejects with explanation of reserved word restriction.	Negative

TC-M12	Browser closed mid-onboarding	Close browser after entering shop name, before submission	Progress saved; merchant resumes from same point on next login.	Edge Case
TC-M13	Second shop creation attempt	Merchant with existing shop attempts to create another	System blocks with V1 limitation message.	Edge Case
TC-M14	Onboarding checklist completion	Merchant completes all 4 onboarding steps	Onboarding marked complete; progress shows 100%; shop is publishable.	Positive

2.1.3 Store Customization (SRS M3)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-M15	Apply new theme	Merchant selects a different theme and saves	Theme applied to live storefront immediately; no downtime for customers.	Positive
TC-M16	Upload logo exceeding 2MB	Image file of 3.5MB uploaded as shop logo	Upload blocked with clear error 'Image must be under 2MB'.	Negative
TC-M17	No logo uploaded	Merchant saves shop without uploading a logo	System displays shop name as text in the header by default.	Edge Case
TC-M18	Customization reflects on live storefront	Merchant saves brand color and tagline changes	Changes visible on live storefront within 5 seconds; customers see updates.	Positive

2.1.4 Product Management (SRS M4)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-M19	Add product — all valid fields	Name, description, price, category, 2 images, 2 variants with stock	Product saved as Active; images on Cloudinary CDN; stock recorded per variant.	Positive
TC-M20	Add product — missing name	Price filled, no name entered	Save blocked; name field highlighted as required.	Negative
TC-M21	Upload image exceeding 2MB	Image file of 3.5MB	Upload blocked client-side; error 'Image must be under 2MB' displayed.	Negative
TC-M22	Variant out of stock after order	Variant stock = 1; customer places order for that variant	Variant marked unavailable automatically; product remains visible with other variants.	Edge Case
TC-M23	Delete product with active orders	Merchant clicks Delete on product linked to a pending order	Warning displayed; deletion blocked; only archiving allowed.	Edge Case
TC-M24	CSV bulk import — valid file (500 products)	Well-formed CSV with 500 product rows	All 500 products imported within 30 seconds; success summary shown.	Performance
TC-M25	CSV bulk import — invalid rows	CSV with 5 rows missing price field	Invalid rows listed with row numbers; valid rows not imported until errors are corrected.	Negative
TC-M26	Duplicate product	Merchant clicks Duplicate on an existing product	New product created with all fields pre-filled; status set to Draft for editing.	Positive
TC-M27	Low-stock threshold alert	Variant stock falls below merchant-defined threshold	Low stock warning appears on merchant dashboard.	Edge Case

2.1.5 Order Management (SRS M5)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-M28	Receive new order notification	Customer places an order on merchant's shop	Merchant receives SMS and email notification within 30 seconds; dashboard badge appears.	Positive
TC-M29	Order status progression	Merchant confirms, then ships, then delivers an order	Status changes: Pending→Confirmed→Shipped→Delivered; customer notified at each change.	Positive
TC-M30	Cancel order with reason	Merchant cancels a Confirmed order with reason 'Out of stock'	Order status updated to Cancelled; customer notified with reason.	Positive
TC-M31	Attempt to cancel Delivered order	Merchant tries to cancel an order already marked Delivered	System blocks cancellation; message directs merchant to resolve directly with customer.	Edge Case
TC-M32	COD order payment collection	Merchant manually marks COD order as 'Payment Collected'	Payment status updated from Pending to Collected; order record updated.	Positive
TC-M33	48-hour inaction reminder	Merchant takes no action on Pending order for 48 hours	System sends reminder notification to merchant.	Edge Case
TC-M34	SMS delivery failure fallback	SMS gateway is unavailable when order is placed	System automatically falls back to email notification for merchant.	Edge Case
TC-M35	Download PDF invoice	Merchant clicks 'Download Invoice' on an order	PDF invoice downloaded containing shop name/logo, order ID, customer details, itemized list, and total.	Positive

2.1.6 Payment Configuration (SRS M6)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-M36	Enable bKash with valid merchant number	Enter valid bKash merchant number and save	bKash enabled; last 4 chars shown; full number masked; bKash appears at customer checkout.	Positive
TC-M37	Invalid SSLCommerz credentials	Enter incorrect Store ID and Store Password for SSLCommerz	System disables card payments and notifies merchant; raw gateway error not exposed.	Negative
TC-M38	All payment methods disabled	Merchant disables all payment methods	System prevents shop from being published; at least one method must be active.	Negative
TC-M39	Disable payment method — existing orders unaffected	Merchant disables bKash; prior orders used bKash	Existing bKash orders remain intact and accessible; change only affects future checkouts.	Edge Case

2.1.7 Analytics & Reporting (SRS M7)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-M40	Analytics dashboard with data	Merchant with 50 orders opens Analytics page	Dashboard loads within 3 seconds; shows revenue, order count, top products, visitor stats.	Positive

TC-M41	Analytics dashboard — zero orders	New merchant with no orders opens Analytics page	Empty state illustrations shown with setup guidance; no blank graphs.	Edge Case
TC-M42	Export CSV report	Merchant selects last 30 days and clicks Export CSV	CSV file downloaded containing order data for the selected date range.	Positive
TC-M43	Cancelled orders in analytics	Merchant views revenue analytics; some orders are cancelled	Cancelled orders excluded from revenue totals; included in order count with 'Cancelled' label.	Edge Case

2.1.8 Account Settings (SRS M8)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-M44	Change password successfully	Enter correct current password and new password meeting requirements	Password updated; all other sessions terminated.	Positive
TC-M45	Enable 2FA via authenticator app	Merchant scans QR code with Google Authenticator and enters TOTP code	2FA enabled; backup codes provided; subsequent logins require TOTP.	Positive
TC-M46	Delete account with active orders	Merchant initiates account deletion with pending orders	System warns to fulfil orders first; deletion not blocked but strongly discouraged.	Edge Case
TC-M47	Google OAuth account — change password option	Google OAuth merchant navigates to Security settings	'Change Password' option hidden; explanatory note shown directing to Google account.	Edge Case

2.2 Customer Tests

2.2.1 Shop Discovery & Browsing (SRS C1)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-C01	Visit published shop via subdomain	Navigate to existingshop.bazarhq.com	Storefront loads in merchant's selected theme within 3 seconds; hero banner and products visible.	Positive
TC-C02	Visit non-existent subdomain	Navigate to fakeshop.bazarhq.com	Friendly 'Shop Not Found' page displayed; no shop data revealed.	Negative
TC-C03	Visit unpublished shop	Navigate to unpublished shop's subdomain	'Shop Currently Unavailable' shown; no shop data or existence confirmed.	Edge Case
TC-C04	Filter products by category	Click 'Fashion' category on storefront	Only products in 'Fashion' category displayed; count shown.	Positive
TC-C05	Search product by keyword	Type 'leather bag' in search bar	Matching products appear within 1 second; results update as customer types.	Positive
TC-C06	Search yields no results	Search for 'xyz123nonexistent'	'No products found' message with suggestion to browse categories.	Negative
TC-C07	View out-of-stock product	Navigate to product detail page with 0 stock	'Add to Cart' button disabled; product clearly marked as out of stock.	Negative
TC-C08	Low stock warning displayed	View product with 3 units remaining	'Only 3 left in stock' warning shown on product detail page.	Edge Case

2.2.2 Cart Management (SRS C2)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-C09	Add in-stock product to cart	Click 'Add to Cart' on available product with variant selected	Item added; cart badge count incremented; confirmation indicator shown.	Positive
TC-C10	Add out-of-stock product to cart	Click 'Add to Cart' on out-of-stock product	Button is disabled; product clearly marked as out of stock.	Negative
TC-C11	Attempt to add quantity exceeding stock	Set quantity to 10 for a product with stock = 5	System prevents quantity above available stock; shows maximum available.	Negative
TC-C12	Cart persists after page refresh	Customer adds items and refreshes the browser	Cart contents remain intact via browser local storage.	Positive
TC-C13	Price change after adding to cart	Product price changes while item is in customer's cart	Cart updates with new price; notice displayed to customer before checkout.	Edge Case
TC-C14	Clear entire cart	Customer clicks 'Clear Cart'	All items removed; cart count shows 0; subtotal resets to zero.	Positive

2.2.3 Checkout & Payment (SRS C3)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-C15	Checkout without account (Guest)	Guest customer fills delivery info and places order	Order placed successfully; no account required; confirmation SMS sent.	Positive
TC-C16	Checkout with invalid phone number	Phone number: 0123456789 (invalid BD format)	Step 1 validation fails; error 'Enter a valid Bangladeshi phone number (01XXXXXXXXXX)'.	Negative
TC-C17	Item stock runs out between cart and checkout	Product in cart; stock set to 0 before checkout	Alert shown at checkout; order placement blocked until item is removed.	Edge Case
TC-C18	SSLCommerz payment — gateway unavailable	Customer selects card payment; SSLCommerz is down	User-friendly error shown; raw gateway errors not exposed; alternative methods suggested.	Edge Case
TC-C19	bKash payment flow	Customer selects bKash, completes payment, enters Transaction ID	Order placed with status 'Payment Pending Verification' until merchant confirms.	Positive
TC-C20	Order confirmation SMS delivery	Customer places an order successfully	SMS with Order ID, items, total, and tracking link delivered within 30 seconds.	Positive
TC-C21	Session expires mid-checkout	Customer's session expires while on Step 2	Cart is preserved; customer redirected to Step 1 upon re-entering checkout.	Edge Case
TC-C22	Review step policies visible	Customer reaches Step 3 — Review	Return policy and shipping policy links are visible on the review page.	Positive

2.2.4 Order Tracking (SRS C4)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-C23	Track order with valid Order ID and phone	Enter correct Order ID and registered phone on /track page	Order tracking page loads; visual timeline shows all stages with timestamps.	Positive
TC-C24	Track order — incorrect Order ID	Enter wrong Order ID	'Order not found' message; system does not confirm whether Order ID exists.	Security
TC-C25	Status update reflected on tracking	Merchant updates order to Shipped	Tracking page reflects 'Shipped' status within 5 seconds.	Positive
TC-C26	Cancelled order on tracking page	View tracking for a cancelled order	'Cancelled' shown with merchant's stated cancellation reason.	Edge Case
TC-C27	Tracking page 90 days after delivery	Customer visits tracking page 91 days post-delivery	Archived message displayed; tracking page no longer shows live data.	Edge Case

2.2.5 Customer Account (SRS C5)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-C28	Create customer account	Register with email and valid password	Account created; verification email sent; customer redirected to account dashboard.	Positive

TC-C29	Save delivery address	Logged-in customer saves a delivery address	Address saved; available for selection at next checkout; pre-fills Step 1.	Positive
TC-C30	Exceed 3 saved addresses limit	Customer with 3 addresses tries to add a 4th	System informs of limit; prompts to delete an existing address first.	Negative
TC-C31	View order history across shops	Logged-in customer opens Order History	All past orders across all BazarHQ shops listed with order ID, shop, date, total, and status.	Positive

2.3 Super Admin Tests

2.3.1 Admin Authentication (SRS A1)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-A01	Admin login — valid credentials + correct 2FA	Valid email, password, and current TOTP code	Admin redirected to platform dashboard; session created (8hr max, 30min inactivity).	Positive
TC-A02	Admin login from non-whitelisted IP	Login attempt from IP not in whitelist	Access denied; blocked page shown with no technical details.	Security
TC-A03	Admin login — 3 consecutive failures	Wrong password submitted 3 times	Account locked for 30 minutes; email alert sent to admin.	Negative
TC-A04	Expired TOTP code entered	Admin enters a TOTP code from previous time-window	2FA step fails; access denied; admin must use current time-window code.	Negative
TC-A05	Admin session auto-termination after inactivity	Admin is idle for 31 minutes	Session terminates automatically; admin redirected to login page.	Edge Case
TC-A06	Admin panel not accessible to public	Attempt to access /superadmin without whitelisted IP	Blocked access page shown; no technical details; admin URL not in search engine index.	Security

2.3.2 Merchant Management (SRS A2)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-A07	Suspend merchant — valid reason provided	Admin selects merchant, enters reason 'Policy Violation', confirms	Merchant sessions terminated; shop unpublished; suspension email sent; audit log updated within 5 seconds.	Positive
TC-A08	Suspend merchant — no reason entered	Admin clicks Suspend but leaves reason field empty	System blocks suspension; prompts admin to enter a reason.	Negative
TC-A09	Admin attempts to view payment credentials	Admin opens merchant detail page and attempts to access gateway credentials	Credentials not displayed; field shows 'Encrypted — Inaccessible'.	Security
TC-A10	Reinstate suspended merchant	Admin clicks Reinstate on a suspended merchant	Merchant receives reinstatement email; can log in and republish shop.	Positive
TC-A11	Suspend merchant with pending orders	Admin suspends merchant who has 3 pending orders	System warns admin about pending orders; orders remain visible on customer tracking.	Edge Case
TC-A12	Search and filter merchant list	Admin searches by shop name 'leather'; filters by Active status	Filtered list shows only matching active merchants; total count updated.	Positive
TC-A13	Permanently delete merchant — confirmation required	Admin initiates permanent deletion without typing confirmation phrase	System blocks deletion; prompts admin to type confirmation phrase.	Security
TC-A14	Admin cannot modify merchant's product or orders	Admin attempts to edit a merchant's product price from detail page	No edit controls available; all merchant data views are read-only for admin.	Security

2.3.3 Platform Analytics & System Health (SRS A3)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-A15	Platform overview dashboard loads	Admin navigates to home dashboard after login	Dashboard loads within 3 seconds; shows total merchants, active shops, today's orders, and revenue.	Positive
TC-A16	System health — service goes down	Database service becomes unavailable	Health monitor reflects outage within 2 minutes; email alert sent to admin.	Edge Case
TC-A17	Analytics export for large date range	Admin exports full platform analytics for 12 months	Export processed as background job; CSV delivered to admin via email when ready.	Performance
TC-A18	Admin attempts to modify analytics data	Admin tries to edit revenue figures in analytics section	No edit controls available; analytics section is strictly read-only.	Security
TC-A19	Audit log integrity	Admin performs merchant suspension; views audit log	Audit log entry created immediately with action type, admin ID, timestamp, and IP address.	Positive
TC-A20	Deleted merchant in platform aggregates	Merchant deletes account; admin views platform revenue	Deleted merchant's historical revenue still included in platform totals; individual record removed.	Edge Case

2.3.4 Theme & Content Management (SRS A4)

Test ID	Test Case	Input / Action	Expected Result	Type
TC-A21	Add and activate new theme	Admin uploads new theme config and sets it as active	Theme appears in merchant selection; preview loads within 3 seconds.	Positive
TC-A22	Delete theme in use by merchants	Admin attempts to delete a theme used by 10 merchants	System blocks deletion; shows count and list of merchants using the theme.	Negative
TC-A23	Send broadcast announcement	Admin composes and sends announcement to all merchants	All merchants receive announcement email within 10 minutes; dashboard notification shown.	Positive
TC-A24	Policy update requires dual approval	Admin updates Privacy Policy and saves	Update queued for second admin approval; not live until approved; updated date stamps on approval.	Security

PART 3 — USER STORIES

User stories capture requirements from each stakeholder's perspective in the format: **As a [role], I want [goal], so that [benefit]**. Each story includes acceptance criteria expressed as Given–When–Then conditions, a priority level (Must Have / Should Have / Could Have), and story point estimates.

Priority Guide:

Must Have — core functionality, MVP blocker

Should Have — important but not MVP-blocking

Could Have — nice-to-have, deferred if needed

3.1 Merchant User Stories

US-M01 **As a new merchant, I want to register on BazarHQ using my email or Google account...** **Must Have | 5 SP pts**

Role: New Merchant As a new merchant, I want to register on BazarHQ using my email or Google account so that I can start building my online shop without technical knowledge.

AC #	Acceptance Criteria (Given – When – Then)
AC-1	Given I am on the registration page, when I enter a unique email and a password with at least 8 characters including one number, then my account is created and I receive a verification email.
AC-2	Given I click 'Register with Google', when I authorize with my Google account, then an account is created automatically without needing email verification.
AC-3	Given I try to register with an already-used email, when I submit the form, then I see 'Email already registered' and no duplicate account is created.
AC-4	Given I have registered but not verified my email, when I try to access the dashboard, then I am redirected to the email verification prompt.

US-M02 **As a merchant, I want to define variants (size, colour) for my products so that ...** **Must Have | 8 SP pts**

Role: Merchant As a merchant, I want to define variants (size, colour) for my products so that customers can choose the specific option they want and stock is tracked accurately per variant.

AC #	Acceptance Criteria (Given – When – Then)
AC-1	Given I am adding a product, when I add a variant type (e.g. Size) with values (S, M, L), then each variant has its own stock quantity and optional price override.
AC-2	Given I define multiple variant types (e.g. Size AND Color), when I save the product, then all combinations are available for selection on the storefront.
AC-3	Given a variant reaches zero stock after an order, when a customer views the product, then that variant is automatically marked unavailable while other variants remain purchasable.
AC-4	Given I set a low-stock threshold, when any variant's stock falls below it, then I see a low-stock warning on my dashboard.

US-M03 **As a merchant, I want to be instantly notified by SMS and email when a new order...** **Must Have | 3 SP pts**

Role: Merchant As a merchant, I want to be instantly notified by SMS and email when a new order arrives so that I can fulfil it promptly and provide good customer service.

AC #	Acceptance Criteria (Given – When – Then)
AC-1	Given a customer places an order, when the order is confirmed by the system, then I receive an SMS and email notification within 30 seconds.
AC-2	Given I have set my preferred notification channel to SMS only, when an order is placed, then only an SMS notification is sent.
AC-3	Given SMS delivery fails, when the system detects the failure, then an email notification is sent automatically as a fallback.
AC-4	Given a pending order has had no action for 48 hours, when this threshold is reached, then the system sends me a reminder notification.

US-M04	As a merchant, I want to view revenue, order, and visitor analytics on my dashbo...	Should Have 5 SP pts
Role: Merchant	As a merchant, I want to view revenue, order, and visitor analytics on my dashboard so that I can understand my shop's performance and make informed business decisions.	
AC #	Acceptance Criteria (Given – When – Then)	
AC-1	Given I open the Analytics page, when the page loads (within 3 seconds), then I see total revenue for today, this week, and this month, plus a 30-day revenue trend chart.	
AC-2	Given I have orders, when I view product analytics, then I see top-selling products ranked by units sold and by revenue generated.	
AC-3	Given I select a custom date range, when I click 'Export CSV', then I download an order or revenue report for that period.	
AC-4	Given I have zero orders, when I open the analytics page, then I see helpful empty-state illustrations with setup guidance instead of blank graphs.	

US-M05	As a merchant, I want to configure which payment methods are available on my sho...	Must Have 5 SP pts
Role: Merchant	As a merchant, I want to configure which payment methods are available on my shop so that customers can pay using the methods most relevant to them in Bangladesh.	
AC #	Acceptance Criteria (Given – When – Then)	
AC-1	Given I navigate to Payment Configuration, when I enable bKash and enter my merchant number, then bKash appears as a payment option for customers at checkout.	
AC-2	Given I enable SSLCommerz and enter valid credentials, when a customer selects card payment, then they are redirected to the SSLCommerz payment page.	
AC-3	Given I try to publish my shop with all payment methods disabled, when I click Publish, then the system prevents publishing and requires at least one active payment method.	
AC-4	Given I mask my payment credentials, when I view the configuration page again, then only the last 4 characters of the credential are shown.	

US-M06	As a merchant, I want to customise my storefront theme, logo, and brand colours ...	Should Have 5 SP pts
Role: Merchant	As a merchant, I want to customise my storefront theme, logo, and brand colours so that my shop looks professional and reflects my brand identity.	
AC #	Acceptance Criteria (Given – When – Then)	
AC-1	Given I navigate to Store Customization, when I select a theme from the 4 available options, then I see a live preview before applying it.	

AC-2	Given I upload a shop logo (PNG/JPG, under 2MB), when I save, then the logo appears in the storefront header immediately.
AC-3	Given I set a primary brand color using the color picker, when I save, then the storefront reflects the new color within 5 seconds.
AC-4	Given I toggle off the hero banner section, when I save, then customers no longer see the hero banner on the homepage.

US-M07	As a merchant, I want to manage my account security and session settings so that...	Must Have 3 SP pts
Role: Merchant	As a merchant, I want to manage my account security and session settings so that my shop and customer data remain protected.	
AC #	Acceptance Criteria (Given – When – Then)	
AC-1	Given I navigate to Security settings, when I enable 2FA with my authenticator app and enter the TOTP code, then 2FA is activated and backup codes are provided.	
AC-2	Given I view my active sessions, when I click 'Terminate' on a session from an unfamiliar device, then that session is immediately invalidated.	
AC-3	Given I request a password reset, when I click the reset link from my email (valid for 1 hour), then I can set a new password and all other sessions are terminated.	
AC-4	Given I want to delete my account, when I type the confirmation phrase and confirm, then my shop is unpublished immediately and full deletion completes within 30 days.	

3.2 Customer User Stories

US-C01

As a guest customer, I want to complete checkout and place an order without crea...

Must Have | 8 SP pts

Role: Guest Customer

As a guest customer, I want to complete checkout and place an order without creating an account so that I can buy products quickly without any registration friction.

AC #	Acceptance Criteria (Given – When – Then)
AC-1	Given I have items in my cart, when I proceed to checkout, then I can complete the full 3-step flow (delivery info, payment, review) without creating an account.
AC-2	Given I enter my delivery details, when I provide my phone number, then the system validates it is a valid Bangladeshi number (01XXXXXXXXXX) before allowing me to proceed.
AC-3	Given I place my order, when the order is confirmed, then I receive a confirmation SMS within 30 seconds containing my Order ID and a tracking link.
AC-4	Given I chose bKash as my payment method, when I submit my Transaction ID, then the order is placed with status 'Payment Pending Verification' until the merchant confirms.

US-C02

As a customer who placed an order without an account, I want to track my order s...

Must Have | 3 SP pts

Role: Guest Customer

As a customer who placed an order without an account, I want to track my order status using my Order ID and phone number so that I always know where my order is.

AC #	Acceptance Criteria (Given – When – Then)
AC-1	Given I am on the tracking page, when I enter my correct Order ID and phone number, then I see a visual timeline showing all order stages with timestamps.
AC-2	Given the merchant updates my order status, when I check the tracking page, then the latest status is reflected within 5 seconds of the merchant saving the change.
AC-3	Given I enter a wrong Order ID or phone number, when I submit the form, then I see 'Order not found' — the system does not confirm whether the Order ID exists.
AC-4	Given my order is cancelled, when I view the tracking page, then I see 'Cancelled' along with the merchant's cancellation reason.

US-C03

As a registered customer, I want to save my delivery addresses to my account so ...

Should Have | 3 SP pts

Role: Registered Customer

As a registered customer, I want to save my delivery addresses to my account so that I can check out faster on future purchases without re-entering my address every time.

AC #	Acceptance Criteria (Given – When – Then)
AC-1	Given I am logged in, when I save a delivery address, then it is stored and can be selected at checkout to pre-fill the delivery information form.
AC-2	Given I have saved 3 addresses (the maximum), when I try to add a 4th, then the system informs me of the limit and prompts me to delete an existing address first.
AC-3	Given I set one address as default, when I initiate checkout, then that address is pre-filled automatically in Step 1.
AC-4	Given I delete a saved address, when I next visit the checkout, then the deleted address is no longer available for selection.

US-C04 As a customer browsing a merchant's storefront, I want to see real-time stock av... Must Have 5 SP pts	
Role: Guest Customer	As a customer browsing a merchant's storefront, I want to see real-time stock availability and variant details so that I only add items to my cart that I can actually purchase.
AC #	Acceptance Criteria (Given – When – Then)
AC-1	Given I view a product detail page, when I select a variant (e.g. Size: L, Color: Blue), then the price and stock status update dynamically to reflect that variant.
AC-2	Given a variant is out of stock, when I view the product, then that variant option is greyed out and I cannot select it or add it to my cart.
AC-3	Given a product has fewer than 5 units remaining, when I view the product detail page, then I see a 'Only X left in stock' warning.
AC-4	Given I add an item to my cart and the stock runs out before checkout, when I proceed to checkout, then I am alerted and prevented from placing the order until the issue is resolved.

US-C05 As a registered customer, I want to view my complete order history across all Ba... Should Have 3 SP pts	
Role: Registered Customer	As a registered customer, I want to view my complete order history across all BazarHQ shops so that I can track past purchases and refer back to any order.
AC #	Acceptance Criteria (Given – When – Then)
AC-1	Given I am logged in, when I navigate to Order History, then I see all past orders across any BazarHQ shop with order ID, shop name, date, items, total, and current status.
AC-2	Given I click on a past order, when the detail page loads, then I see the full order details and can access the tracking page directly from my account.
AC-3	Given I place a guest order with my registered email address, when the order is placed, then the order is automatically associated with my account and visible in Order History.
AC-4	Given I delete my account, when deletion is confirmed, then I lose access to order history on the platform, but merchants retain their order records.

3.3 Super Admin User Stories

US-A01

As a Super Admin, I want to log in using email, password, and a TOTP 2FA code so...

Must Have | 5 SP pts

Role: Super Admin

As a Super Admin, I want to log in using email, password, and a TOTP 2FA code so that the admin panel is protected against unauthorized access even if my password is compromised.

AC #	Acceptance Criteria (Given – When – Then)
AC-1	Given I navigate to bazarhq.com/superadmin from a whitelisted IP, when I enter valid credentials, then I am prompted for a TOTP code before gaining access.
AC-2	Given I enter 3 consecutive wrong passwords, when the 3rd failure occurs, then my account is locked for 30 minutes and I receive an email alert.
AC-3	Given my session has been idle for 30 minutes, when I attempt to take an action, then I am redirected to the login page.
AC-4	Given I log in successfully, when I perform any admin action, then that action is recorded in the immutable audit log with my ID, timestamp, and IP address.

US-A02

As a Super Admin, I want to suspend a merchant account with a stated reason so t...

Must Have | 5 SP pts

Role: Super Admin

As a Super Admin, I want to suspend a merchant account with a stated reason so that policy-violating merchants are immediately removed from the platform while merchants understand why they were suspended.

AC #	Acceptance Criteria (Given – When – Then)
AC-1	Given I open a merchant's detail page, when I click 'Suspend Account', then I am required to enter a reason before the action proceeds.
AC-2	Given I confirm a suspension with a reason, when the action completes (within 5 seconds), then the merchant's sessions are terminated, their shop is unpublished, and they receive an email with the stated reason.
AC-3	Given a suspended merchant has pending orders, when I suspend them, then the system warns me and those orders remain visible to customers on tracking pages.
AC-4	Given I reinstate a suspended merchant, when the action completes, then the merchant receives a reinstatement email and can log in to republish their shop.

US-A03

As a Super Admin, I want to see the operational status of all critical services ...

Should Have | 3 SP pts

Role: Super Admin

As a Super Admin, I want to see the operational status of all critical services in real time so that I can identify and respond to outages before they impact merchants and customers.

AC #	Acceptance Criteria (Given – When – Then)
AC-1	Given I navigate to System Health, when the page loads, then I see the current status (Up/Down) of the web server, database, storage, email gateway, and SMS gateway.
AC-2	Given any critical service goes down, when the monitor detects it (within 60 seconds), then I receive an email alert automatically.
AC-3	Given I view the error log, when the page loads, then I see all system errors from the past 7 days with timestamps.
AC-4	Given I am on the System Health page, when a service status changes, then the page reflects the new status within 2 minutes.

US-A04		As a Super Admin, I want to send broadcast announcements to all merchants so tha...	Should Have 3 SP pts
Role: Super Admin		As a Super Admin, I want to send broadcast announcements to all merchants so that I can communicate platform updates, policy changes, and important notices efficiently.	
AC #	Acceptance Criteria (Given – When – Then)		
AC-1	Given I compose a broadcast announcement, when I click Send, then all registered merchants receive the announcement via email within 10 minutes and see a dashboard notification banner.		
AC-2	Given I want to schedule an announcement, when I set a future date and time and save, then the announcement is sent automatically at the scheduled time.		
AC-3	Given I am about to send an announcement, when I click Send, then a confirmation screen shows the recipient count before I can confirm; the announcement cannot be recalled after sending.		
AC-4	Given I want to update the Terms of Service, when I save the changes, then the update is queued for a second admin's approval before going live with an updated date stamp.		

DOCUMENT SUMMARY & SIGN-OFF

Metric	Count / Value
Total Use Case Scenarios	12 (UC-M01–UC-M08, UC-C01–UC-C04, UC-A01–UC-A04)
Total Test Cases	72+ (Merchant 47, Customer 22, Super Admin 18)
Total User Stories	16 (Merchant 7, Customer 5, Super Admin 4)
Total Acceptance Criteria	64 (avg 4 per story)
SRS Modules Covered	All 17 modules (M1–M8, C1–C5, A1–A4)
Test Case Types	Positive, Negative, Edge Case, Security, Performance
Priority Distribution	Must Have: 10 stories Should Have: 6 stories
Total Story Points	~65 SP across all user stories
Document Version	1.0 — April 2026
Prepared By	BazarHQ Development Team
Based On	BazarHQ SRS v1.0 (IEEE 830)

Note: This document is maintained by the BazarHQ Development Team. All scenarios are derived from and traceable to the BazarHQ SRS v1.0. Any updates to the SRS must be reflected in this document. All scenarios have been reviewed against IEEE 830 requirements engineering standards.